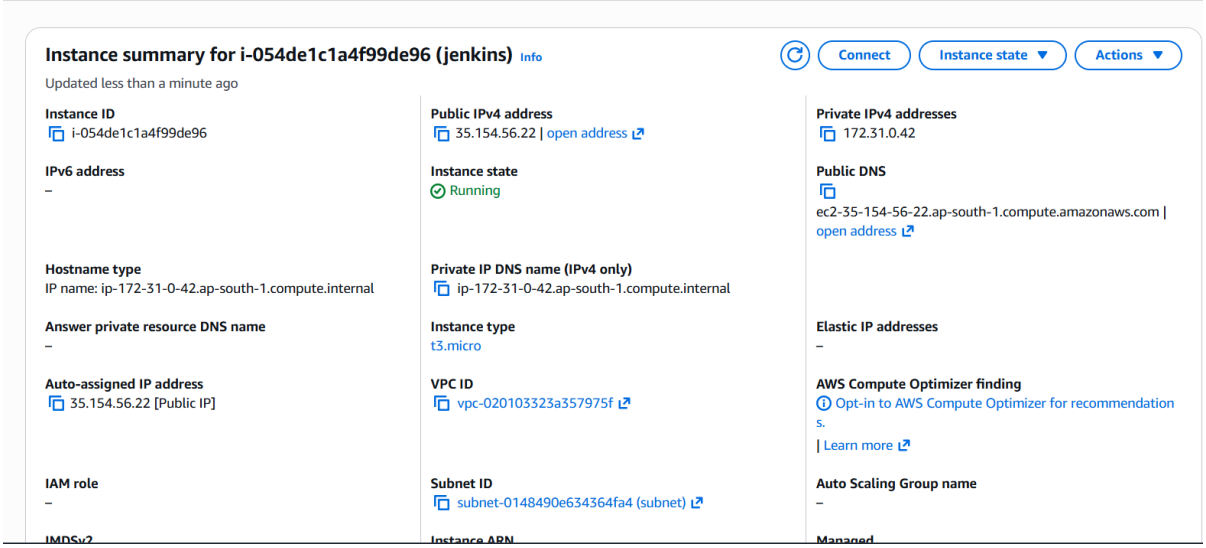
Created an Ec2 instance named Jenkins



From ec2 instance connected to Ubuntu server by using ssh

```
sravanth@NANI:~$ ssh -i "aws-key.pem" ubuntu@ec2-35-154-56-22.ap-south-1.compute.amazonaws.com
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1011-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Aug 19 16:25:16 UTC 2026

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   59.4% of 6.61GB   Processes:             115
  Memory usage: 73%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.0.42

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

   https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Wed Aug 19 11:49:27 2026 from 183.82.96.78
```

Installed Java21

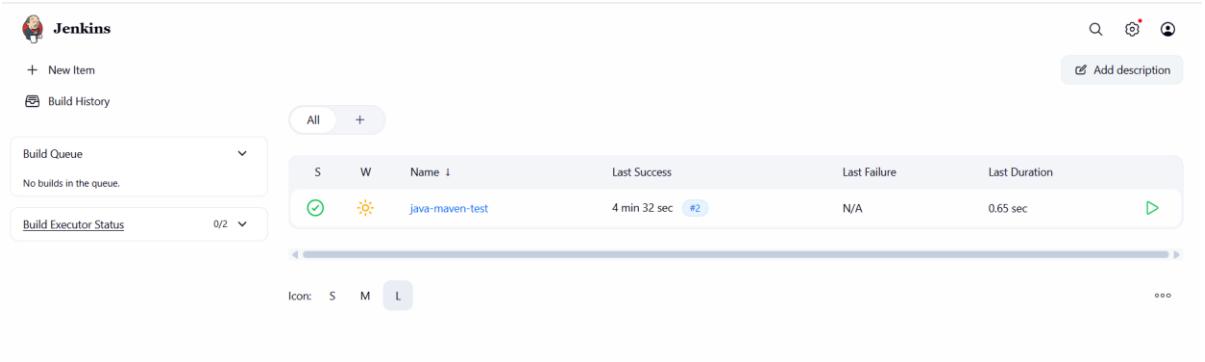
Installed the maven and installed Jenkins and checked and checked whether Jenkins is running or not

```
ubuntu@ip-172-31-0-42:~$ sudo systemctl enable jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
ubuntu@ip-172-31-0-42:~$ sudo systemctl start jenkins
ubuntu@ip-172-31-0-42:~$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-08-19 04:58:04 UTC; 1min 14s ago
 Invocation: ab51a76b181f46daa2fb1d7fbe1bf222
   Main PID: 3914 (java)
      Tasks: 37 (limit: 627)
     Memory: 312.3M (peak: 352.2M)
        CPU: 20.078s
     CGroup: /system.slice/jenkins.service
             └─3914 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Aug 19 04:57:57 ip-172-31-0-42 jenkins[3914]: [LF]>
Aug 19 04:57:57 ip-172-31-0-42 jenkins[3914]: [LF]> ************************************************************
Aug 19 04:57:57 ip-172-31-0-42 jenkins[3914]: [LF]> ************************************************************
Aug 19 04:57:57 ip-172-31-0-42 jenkins[3914]: [LF]> ************************************************************
Aug 19 04:58:04 ip-172-31-0-42 jenkins[3914]: 2026-08-19 04:58:04.324+0000 [id=34]        INFO        jenkins.InitReactorRunner$1#onAttained: Completed ini>
Aug 19 04:58:04 ip-172-31-0-42 jenkins[3914]: 2026-08-19 04:58:04.348+0000 [id=24]        INFO        hudson.lifecycle.Lifecycle#onReady: Jenkins is fully >
Aug 19 04:58:04 ip-172-31-0-42 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Aug 19 04:58:05 ip-172-31-0-42 jenkins[3914]: 2026-08-19 04:58:05.945+0000 [id=50]        INFO        h.m.DownloadService$Downloadable#load: Obtained the u>
Aug 19 04:58:05 ip-172-31-0-42 jenkins[3914]: 2026-08-19 04:58:05.945+0000 [id=50]        INFO        hudson.util.Retrier#start: Performed the action check>
Aug 19 04:58:09 ip-172-31-0-42 jenkins[3914]: 2026-08-19 04:58:09.446+0000 [id=69]        WARNING        h.n.DiskSpaceMonitorDescriptor#markNodeOfflineOrOn>

ubuntu@ip-172-31-0-42:~$ /var/lib/jenkins/secrets/initialAdminPassword
```

Verified required plugins executed and also verified Jenkins can execute maven and created a free style java project



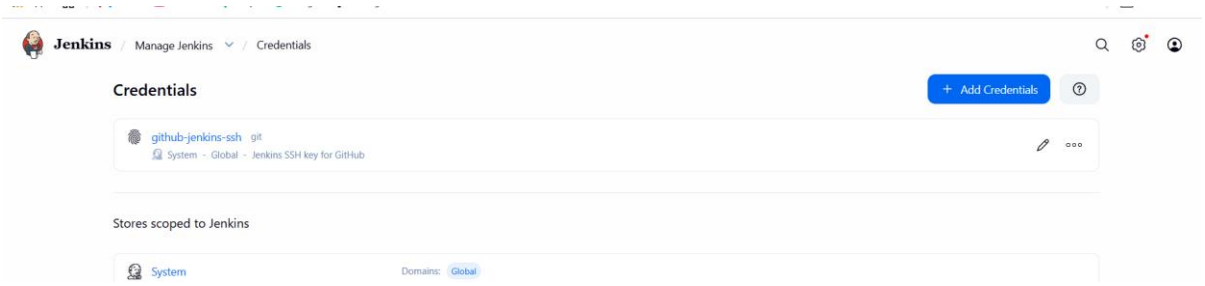
I created a ssh key-pair on jenkins server it will create public and private keys

I added that public key in the github repository deploy keys

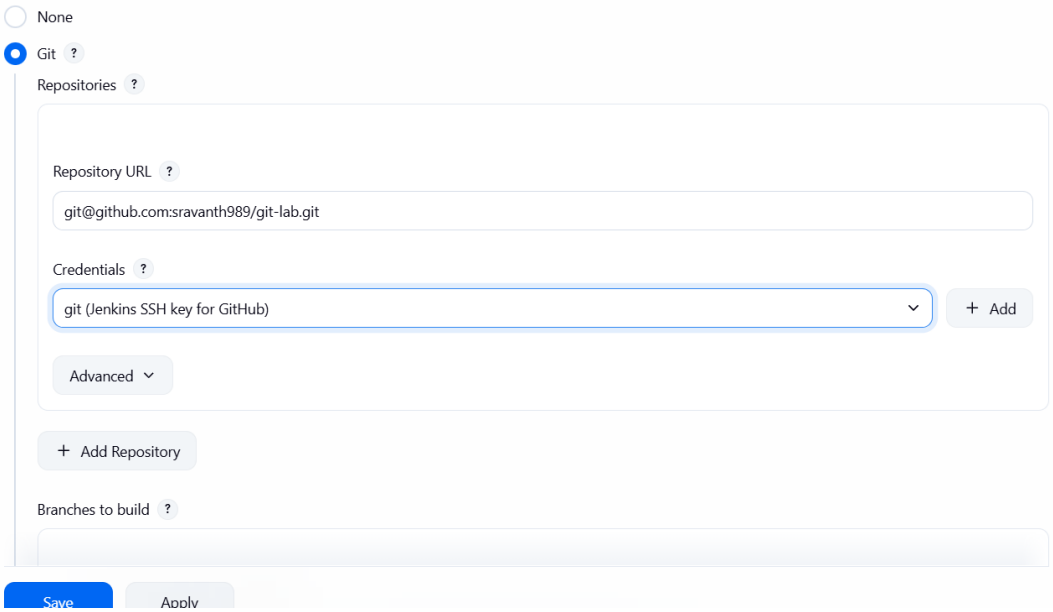
After added I tested

```
ubuntu@ip-172-31-0-42:~$ sudo -u jenkins ssh -T git@github.com
Hi sravanth989/git-lab! You've successfully authenticated, but GitHub does not provide shell access.
ubuntu@ip-172-31-0-42:~$
```

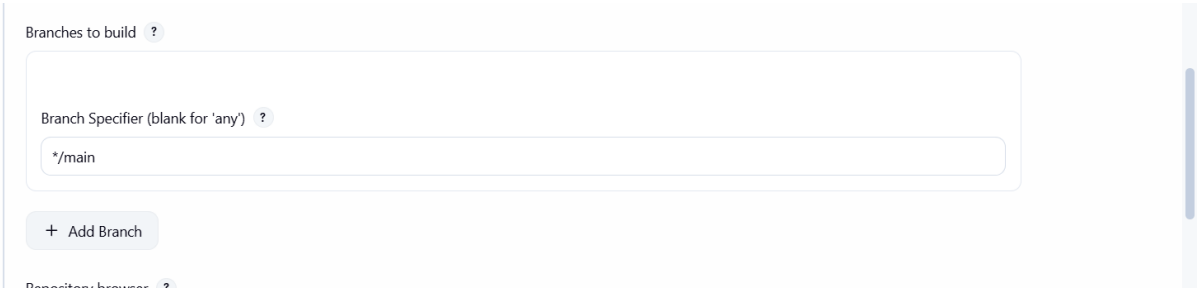
Store the private key in Jenkins credentials



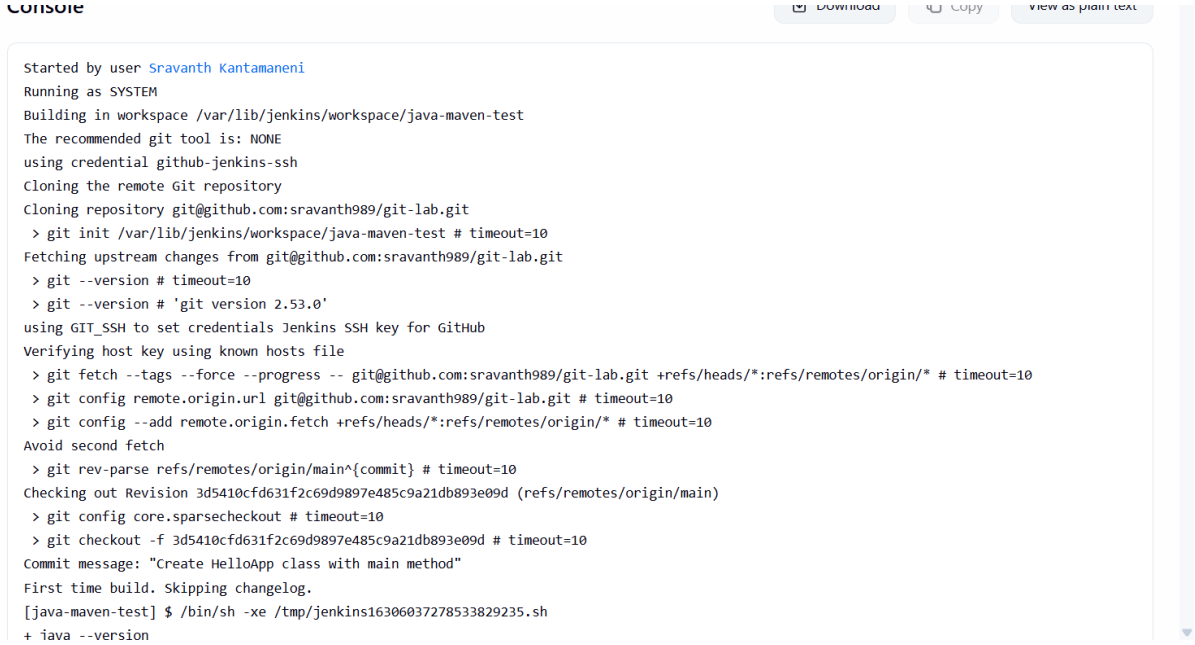
After this go to java-maven-test and next to configure on that go to source code management and add git . It shows that the Jenkins job is configured to use the GitHub repository and SSH credential



In the same section branch configuration



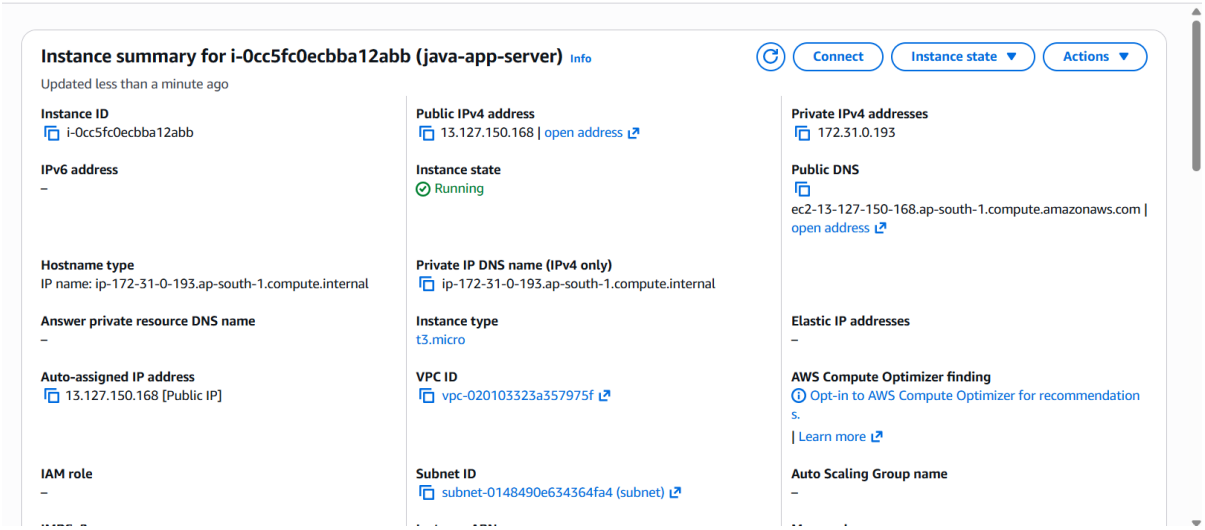
After clicking save Jenkins successfully authenticated with GitHub and checked out the main branch.



```
1.18.jar[90m (592 kB at 1.8 MB/s)[0m
Progress (1): 233/263 kB
Progress (1): 237/263 kB
Progress (1): 241/263 kB
Progress (1): 245/263 kB
Progress (1): 249/263 kB
Progress (1): 253/263 kB
Progress (1): 257/263 kB
Progress (1): 261/263 kB
Progress (1): 263 kB

Downloaded[90m from [0mcentral[90m: https://repo.maven.apache.org/maven2/[0morg/codehaus/plexus/plexus-utils/3.2.0/plexus-utils-
3.2.0.jar[90m (263 kB at 744 kB/s)[0m
[[1;34mINFO[m] Building jar: /var/lib/jenkins/workspace/java-maven-test/target/hello-app-1.0.jar
[[1;34mINFO[m] [1m------------------------------------------------------------------------[m
[[1;34mINFO[m] [1;32mBUILD SUCCESS[m
[[1;34mINFO[m] [1m------------------------------------------------------------------------[m
[[1;34mINFO[m] Total time:  13.941 s
[[1;34mINFO[m] Finished at: 2026-08-20T06:10:54Z
[[1;34mINFO[m] [1m------------------------------------------------------------------------[m
Finished: SUCCESS
```

Created Application EC2 Instance



Connecting java ec2 server to Ubuntu by using ssh

And installed java on java server

```
ubuntu@ip-172-31-0-193:~$ java --version
openjdk 21.0.11 2026-04-21
OpenJDK Runtime Environment (build 21.0.11+10-1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.11+10-1-26.04.2-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-0-193:~$
```

Java Application Deployment Directory Created

```
ubuntu@ip-172-31-0-193:~$ sudo mkdir -p /opt/java-app
ubuntu@ip-172-31-0-193:~$ sudo chown ubuntu:ubuntu /opt/java-app
ubuntu@ip-172-31-0-193:~$ ls -ld /opt/java-app
drwxr-xr-x 2 ubuntu ubuntu 4096 Aug 20 06:56 /opt/java-app
ubuntu@ip-172-31-0-193:~$
```

Jenkins Public Key Added to Application EC2

```
ubuntu@ip-172-31-0-193: $ ls -ld ~/.ssh
ls -l ~/.ssh/authorized_keys
drwx------ 2 ubuntu ubuntu 4096 Aug 20 07:03 /home/ubuntu/.ssh
-rw------- 1 ubuntu ubuntu 96 Aug 20 07:03 /home/ubuntu/.ssh/authorized_keys
ubuntu@ip-172-31-0-193: $
```

Jenkins Successfully Connected to Application EC2

```
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Aug 20 07:06:11 UTC 2026

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   42.2% of 6.61GB   Processes:             117
  Memory usage: 29%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.0.193

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

   https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

114 updates can be applied immediately.
83 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Thu Aug 20 06:48:59 2026 from 183.82.96.78
ubuntu@ip-172-31-0-193:~$
```

Application Deployment Directory Created

```
ubuntu@ip-172-31-0-193:~$ java -version
openjdk version "21.0.11" 2026-04-21
OpenJDK Runtime Environment (build 21.0.11+10-1~26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.11+10-1~26.04.2-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-0-193:~$ ls -la
total 32
drwxr-x--- 5 ubuntu ubuntu 4096 Aug 20 07:01 .
drwxr-xr-x 3 root   root   4096 Aug 20 06:44 ..
-rw-r--r-- 1 ubuntu ubuntu  220 Feb 13  2026 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Feb 13  2026 .bashrc
drwx------ 2 ubuntu ubuntu 4096 Aug 20 06:48 .cache
drwxrwxr-x 3 ubuntu ubuntu 4096 Aug 20 07:01 .local
-rw-r--r-- 1 ubuntu ubuntu  807 Feb 13  2026 .profile
drwx------ 2 ubuntu ubuntu 4096 Aug 20 07:03 .ssh
ubuntu@ip-172-31-0-193:~$
```

JAR Successfully Copied from Jenkins to Application EC2

```
ubuntu@ip-172-31-0-42:~$ sudo -u jenkins scp /var/lib/jenkins/workspace/java-maven-test/target/*.jar ubuntu@13.127.150.1
68:/home/ubuntu/app/
hello-app-1.0.jar                                             100% 1791     1.5MB/s   00:00
ubuntu@ip-172-31-0-42:~$
```

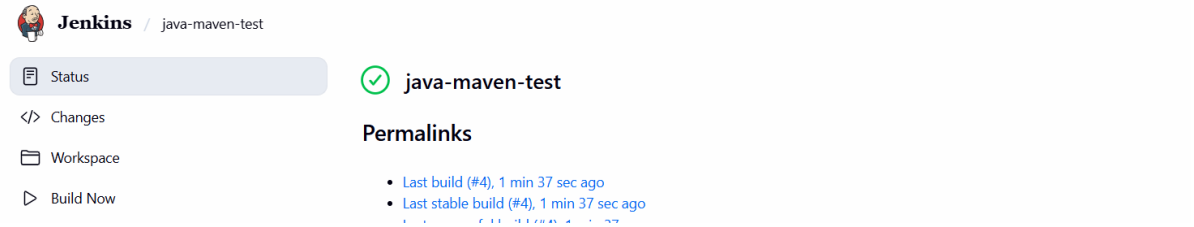
AR Verified on Application EC2

```
ubuntu@ip-172-31-0-42:~$ sudo -u jenkins ssh ubuntu@13.127.150.168 "ls -lh /home/ubuntu/app/"
total 4.0K
-rw-r--r-- 1 ubuntu ubuntu 1.8K Aug 20 07:14 hello-app-1.0.jar
ubuntu@ip-172-31-0-42:~$
```

Maven JAR Manifest Configuration Added

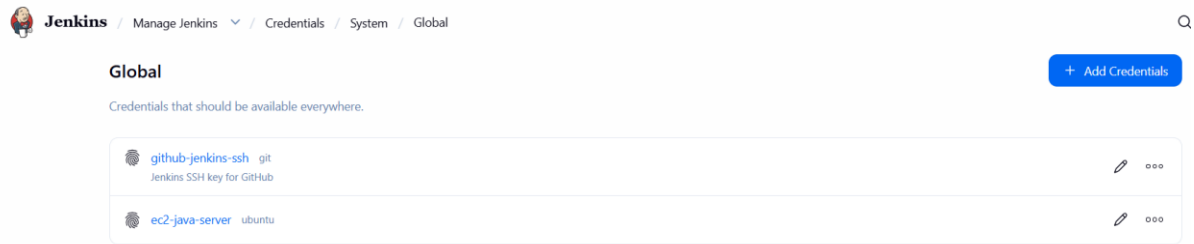


Jenkins Build Successful After Manifest Fix

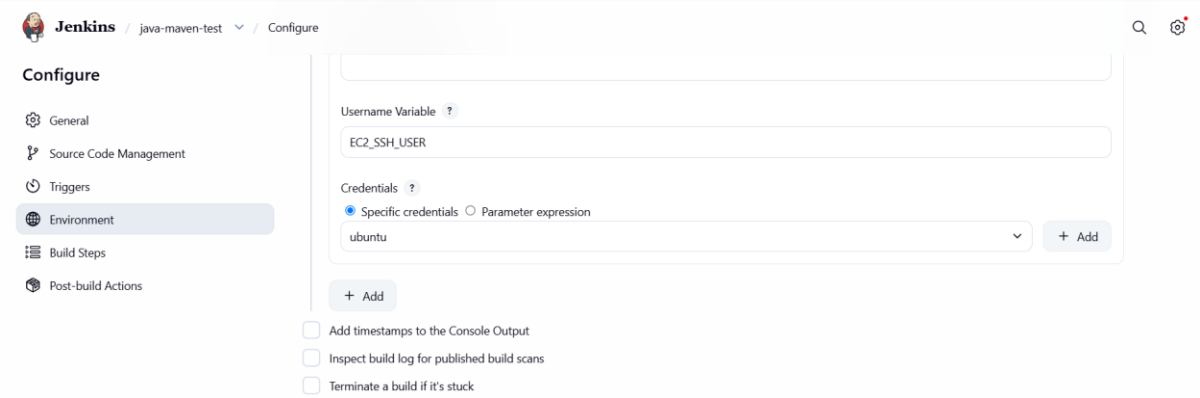


Jenkins Secret Binding configured and saved

Create Jenkins Credential for Java EC2



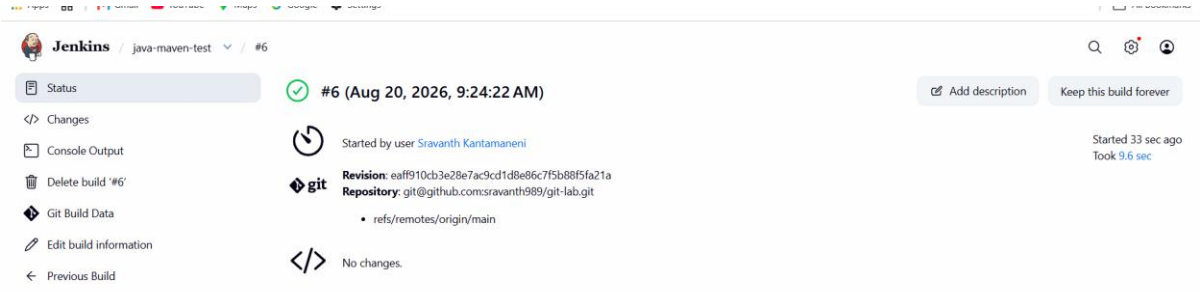
EC2 SSH Credential Bound to Jenkins Environment



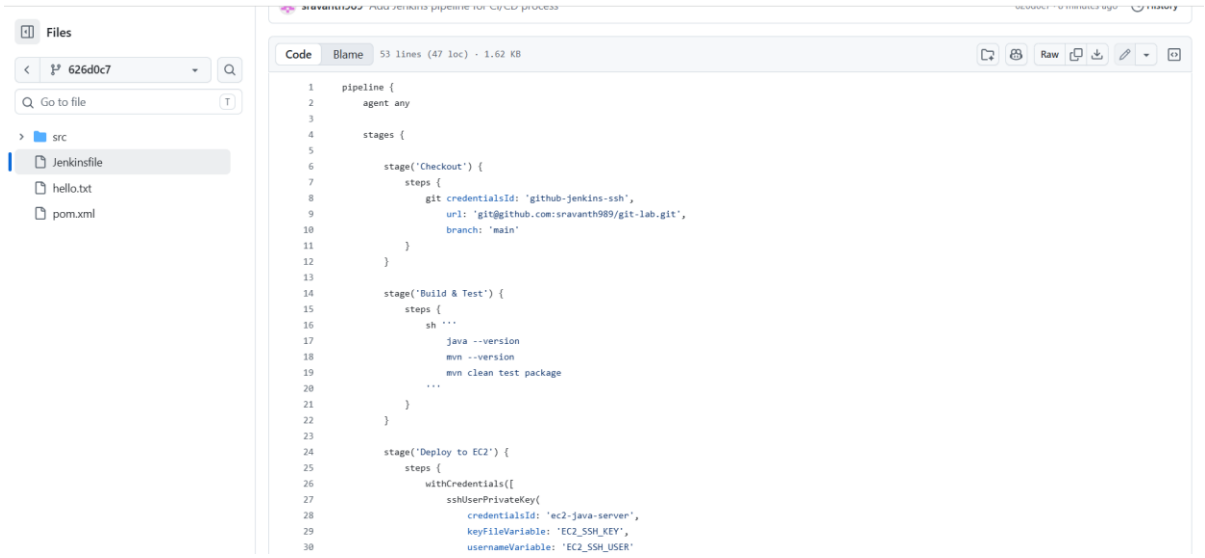
Showing the Deployment Execute shell.



Successful Jenkins Build and Java EC2 Deployment



Creating Jenkinsfile in GitHub and commited



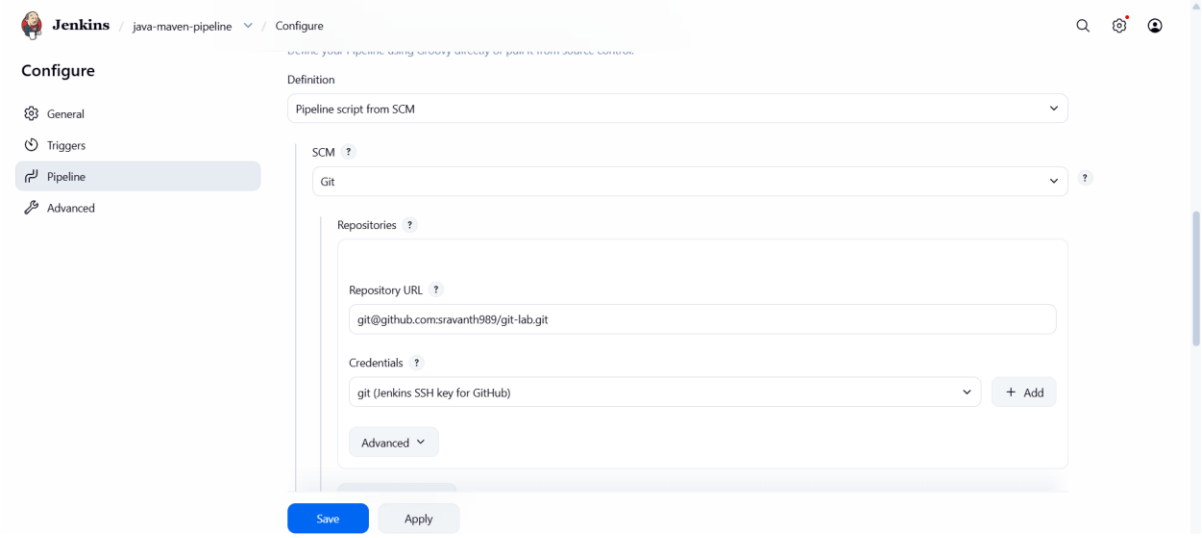

```
        stage('Deploy to EC2') {
            steps {
                withCredentials([
                    sshUserPrivateKey(
                        credentialsId: 'ec2-java-server',
                        keyFileVariable: 'EC2_SSH_KEY',
                        usernameVariable: 'EC2_SSH_USER'
                    )
                ]) {
                    sh '''
                        scp -i "$EC2_SSH_KEY" \
                        -o StrictHostKeyChecking=no \
                        target/*.jar \
                        "$EC2_SSH_USER"@13.127.150.168:/home/ubuntu/app/

                        ssh -i "$EC2_SSH_KEY" \
                        -o StrictHostKeyChecking=no \
                        "$EC2_SSH_USER"@13.127.150.168 \
                        "pkill -f '[h]ello-app-1.0.jar' || true"

                        ssh -i "$EC2_SSH_KEY" \
                        -o StrictHostKeyChecking=no \
                        "$EC2_SSH_USER"@13.127.150.168 \
                        "nohup java -jar /home/ubuntu/app/hello-app-1.0.jar > /home/ubuntu/app/app.log 2>&1 &"
                    '''
                }
            }
        }
    }
}
```

After that go to Jenkins dashboard and new item and select pipeline

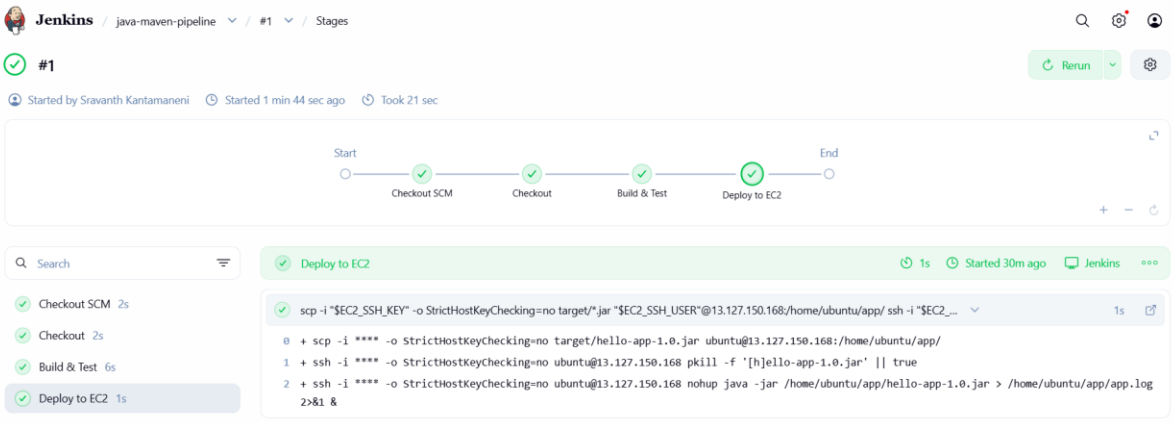
Now after creating a pipeline we can the scm from git and give the repository URL and give the credentials after entering click save.



Run the pipeline by clicking Build now

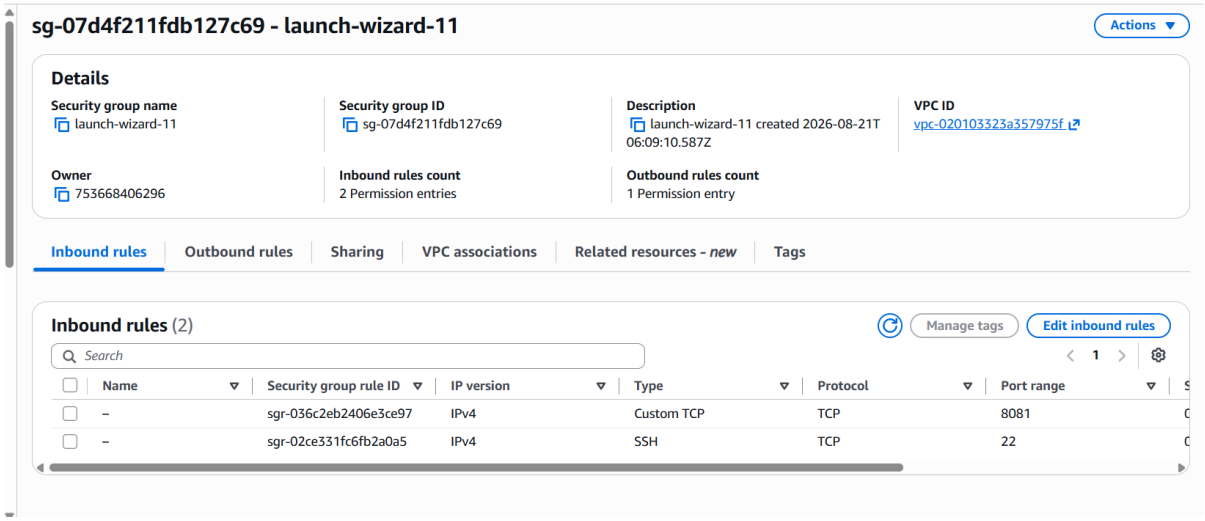
Jenkins will checkout code from Github

And build and test using maven and it gets success.

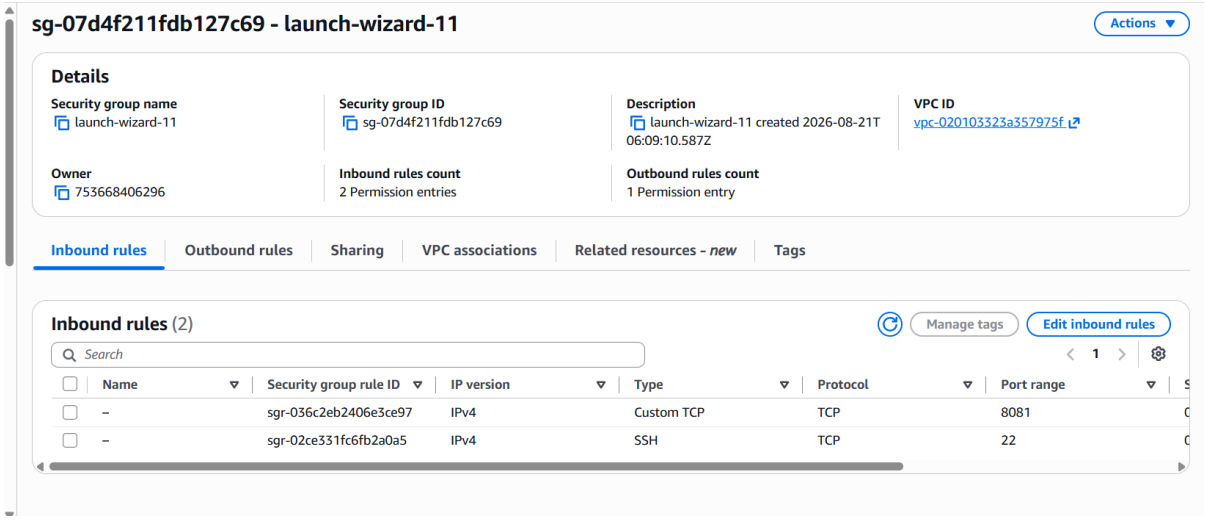


Install/configure Nexus, create a Maven repository, and publish the versioned JAR from Jenkins to Nexus.

Created a ec2 instance for nexus



Configured the Nexus EC2 Security Group to allow SSH (22) and Nexus UI access on port 8081.



Connected to nexus server using ssh

```
System information as of Fri Aug 21 06:35:58 UTC 2026

 System load:  0.0                Temperature:           -273.1 C
 Usage of /:   11.1% of 18.25GB   Processes:             114
 Memory usage: 13%                Users logged in:       0
 Swap usage:   0%                 IPv4 address for ens5: 172.31.0.129

xpanded Security Maintenance for Applications is not enabled.

 updates can be applied immediately.

nable ESM Apps to receive additional future security updates.
ee https://ubuntu.com/esm or run: sudo pro status


he list of available updates is more than a week old.
o check for new updates run: sudo apt update


he programs included with the Ubuntu system are free software;
he exact distribution terms for each program are described in the
ndividual files in /usr/share/doc/*/copyright.

buntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
pplicable law.

buntu@ip-172-31-0-129:~$
```

Installed java on nexus server

```
ubuntu@ip-172-31-0-129:~$ java -version
openjdk version "21.0.11" 2026-04-21
OpenJDK Runtime Environment (build 21.0.11+10-1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.11+10-1-26.04.2-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-0-129:~$
```

Verified that Docker Engine is installed and the Docker service is running successfully.

```
ubuntu@ip-172-31-0-129:/opt$ sudo systemctl status docker --no-pager
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Fri 2026-08-21 09:22:03 UTC; 58s ago
 Invocation: 2ee8db1c2a924578bf2cf458247e4669
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 9131 (dockerd)
      Tasks: 9
     Memory: 28.4M (peak: 28.8M)
        CPU: 256ms
     CGroup: /system.slice/docker.service
             └─9131 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Aug 21 09:22:03 ip-172-31-0-129 dockerd[9131]: time="2026-08-21T09:22:03.012638110Z" level=info msg="Restoring c…start."
Aug 21 09:22:03 ip-172-31-0-129 dockerd[9131]: time="2026-08-21T09:22:03.049351005Z" level=info msg="Deleting nf…atus 1"
Aug 21 09:22:03 ip-172-31-0-129 dockerd[9131]: time="2026-08-21T09:22:03.053808350Z" level=info msg="Deleting nf…atus 1"
Aug 21 09:22:03 ip-172-31-0-129 dockerd[9131]: time="2026-08-21T09:22:03.243618355Z" level=info msg="Loading con… done."
Aug 21 09:22:03 ip-172-31-0-129 dockerd[9131]: time="2026-08-21T09:22:03.248559027Z" level=info msg="Docker daem…=29.7.2
Aug 21 09:22:03 ip-172-31-0-129 dockerd[9131]: time="2026-08-21T09:22:03.248655925Z" level=info msg="Initializin…ildkit"
Aug 21 09:22:03 ip-172-31-0-129 dockerd[9131]: time="2026-08-21T09:22:03.322898583Z" level=info msg="Completed b…zation"
Aug 21 09:22:03 ip-172-31-0-129 dockerd[9131]: time="2026-08-21T09:22:03.328879374Z" level=info msg="Daemon has …zation"
Aug 21 09:22:03 ip-172-31-0-129 dockerd[9131]: time="2026-08-21T09:22:03.328919818Z" level=info msg="API listen …r.sock"
Aug 21 09:22:03 ip-172-31-0-129 systemd[1]: Started docker.service - Docker Application Container Engine.
Hint: Some lines were ellipsized, use -l to show in full.
ubuntu@ip-172-31-0-129:/opt$
```

Created a persistent Docker volume to store Nexus Repository data.

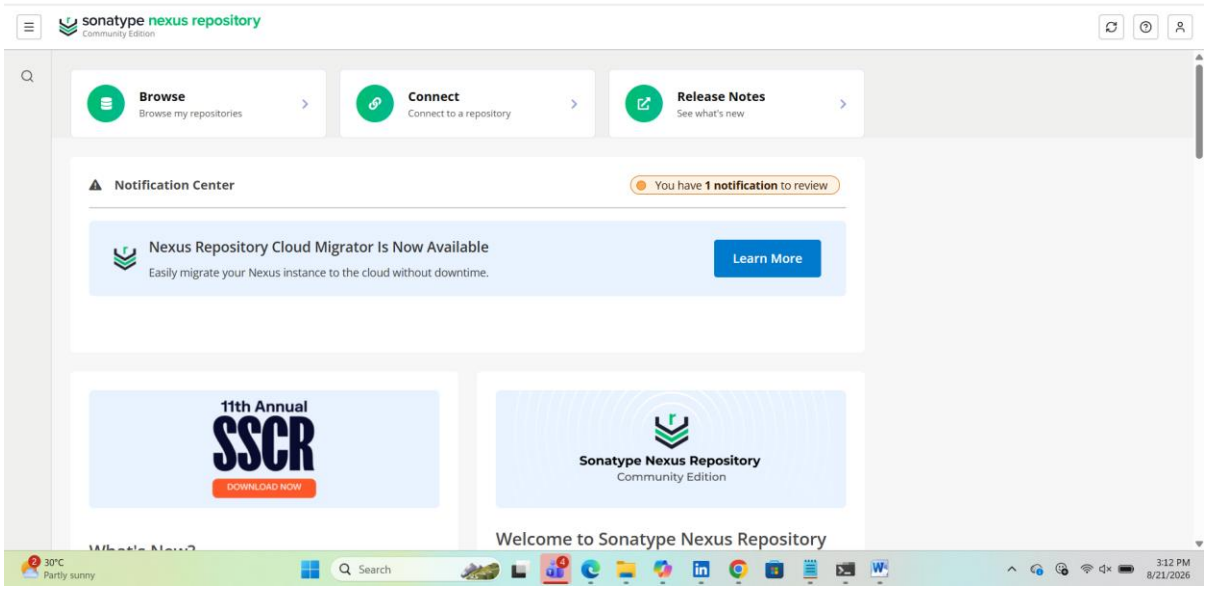
```
ubuntu@ip-172-31-0-129:/opt$ sudo docker volume create nexus-data
nexus-data
ubuntu@ip-172-31-0-129:/opt$ sudo docker volume ls
DRIVER    VOLUME NAME
local     nexus-data
ubuntu@ip-172-31-0-129:/opt$
```

Start nexus using docker

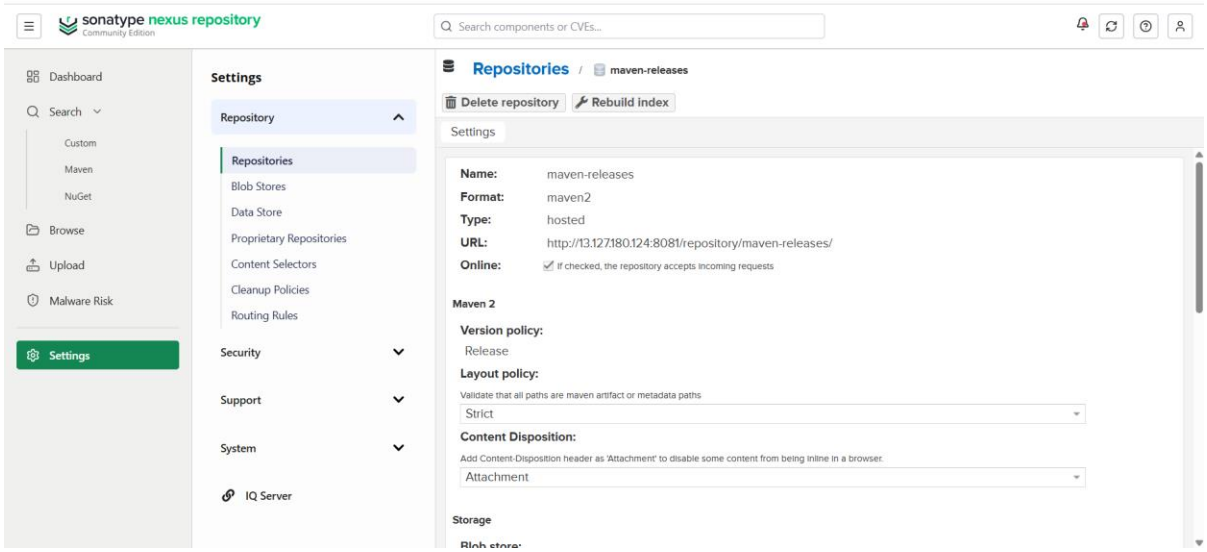
Nexus Repository as a Docker container, exposes its web interface on port 8081, and connects persistent storage for Nexus data.

```
ubuntu@ip-172-31-0-129:/opt$ sudo docker run -d \
  --name nexus \
  -p 8081:8081 \
  -v nexus-data:/nexus-data \
  sonatype/nexus3
Unable to find image 'sonatype/nexus3:latest' locally
latest: Pulling from sonatype/nexus3
80db3c7b682f: Pull complete
55afa1ecc21d: Pull complete
75086f395fc0: Pull complete
b76c1e4bcc19: Pull complete
90d263d7dc81: Pull complete
249fa7c8aeff: Pull complete
e9513064123d: Pull complete
3cd84616f714: Pull complete
44136fa355b3: Download complete
c5c48429e4c2: Download complete
Digest: sha256:1263e77bd1e7748d3be9028ba6472cca7f534828e92d157cd0f3c13c6909b150
Status: Downloaded newer image for sonatype/nexus3:latest
5da7e04569d553bd35fdfb3e431cc4ee6d8ed733ec15804a842e0cbbc795d096
ubuntu@ip-172-31-0-129:/opt$ sudo docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED              STATUS              PORTS
                    NAMES
5da7e04569d5   sonatype/nexus3   "/opt/sonatype/nexus…"   About a minute ago   Up About a minute   0.0.0.0:8081->8081/tc
p, [::]:8081->8081/tcp   nexus
ubuntu@ip-172-31-0-129:/opt$
```

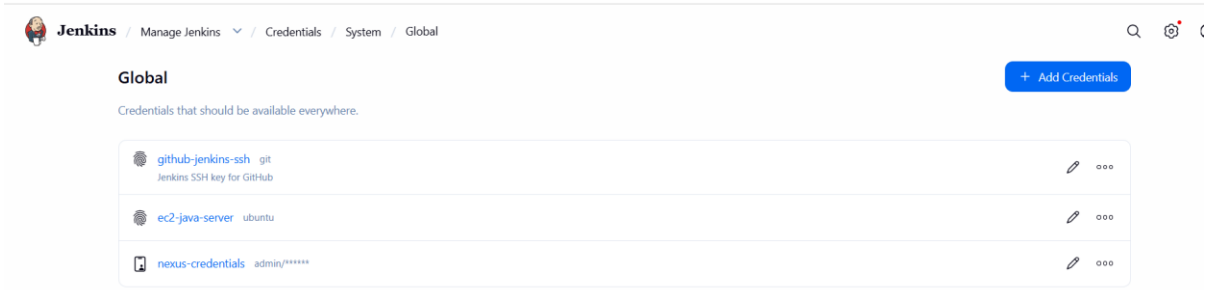
Accessed the Nexus Repository web interface through port 8081.



Created a Maven hosted repository named `maven-releases` in Nexus to store release JAR artifacts.



Placed nexus credentials in Jenkins



I updated Jenkins file locally added this commands

```
        }

        stage('Publish to Nexus') {
            steps {
                withCredentials([usernamePassword(
                    credentialsId: 'nexus-credentials',
                    usernameVariable: 'NEXUS_USER',
                    passwordVariable: 'NEXUS_PASSWORD'
                )]) {
                    sh '''
                        mvn deploy \
                        -DaltDeploymentRepository=nexus::default::http://13.127.180.124:8081/repository/maven-releases/
\
                        -Dusername="$NEXUS_USER" \
                        -Dpassword="$NEXUS_PASSWORD"
                    '''
                }
            }
```

Now go to Jenkins and run the pipeline

Clicked buildnow

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